Shenghan **Gao**

Optics · Photonics · Computer Science

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Summary

Integrated photonics and optical systems researcher combining chip design, optical characterization, signal processing, and Python-based lab automation. Led experimental work on injection-locked coherent photonic networks, built reusable instrument-control workflows, and connected device-level measurements to system performance. Strong fit for photonics R&D, optical sensing, silicon photonics, and hardware/software measurement roles.

Core Skills

Photonics	Integrated photonics, optical injection locking, coherent optical systems, fiber-optic sensing, FBG/BOTDR, optical characterization
Lab systems	Lasers, modulators, free-space/fiber optics, OSAs, SVAs, lock-in amplifiers, photodetectors, e-beam lithography, SEM
Automation	Python, NumPy, SciPy, Pandas, Matplotlib, PyVISA/SCPI, NI MAX, MATLAB, C++, Git/GitHub, pytest
Simulation	Ansys Lumerical, COMSOL, Zemax OpticStudio, CodeV, RSOFT, MEEP, RCWA, LTspice, SolidWorks, NX
ML / CV	scikit-learn, PyTorch, TensorFlow, OpenCV, CNN image recognition
Languages	Chinese (Native), English (Fluent), Japanese (Fluent)

Research Experience

University of Pittsburgh

Pittsburgh, PA

Research Assistant | Advisor: Nathan Youngblood

Spring 2020 - present

- Led remote optical injection-locking experiments between long-haul master and local slave lasers, establishing phase-coherent operation for coherent detection, interference-based signal processing, and optical-domain computing.
- Designed photonic waveguides/components in KLayout and validated fabricated chips through optical characterization of coupling loss, interference contrast, and modulator response.
- Modeled device-to-system behavior in Ansys Lumerical, mapping modulator efficiency, insertion loss, and coupling loss to locking range, SNR, and bandwidth.
- Created object-oriented Python automation with NI MAX and PyVISA/SCPI for lasers, SVAs, OSAs, and related instruments, turning manual lab routines into reusable sweeps, batch acquisition, and standardized analysis pipelines.
- Built FBG/BOTDR sensing workflows, including tapered-fiber fabrication, COMSOL thermo-mechanical modeling, temperature-wavelength calibration, sensitivity extraction, drift compensation, and uncertainty analysis.

University of Rochester

Rochester, NY

Research Assistant | Advisor: Xi-Cheng Zhang

Sep 2017 - Aug 2019

- Generated THz signals from liquid plasmas and analyzed how temperature, polarity, and viscosity affected emission strength and spectral content.
- Processed time- and frequency-domain experimental data with SNR, repeatability, sensitivity, and regression analysis; also prototyped digital cloaking using lenslet arrays and MATLAB pixel remapping.

Industry Experience

Lumentum Operations LLC

San Jose, CA

Research Intern | Mentor: Joe Wen

May 2023 - Aug 2023

- Built MATLAB/Python workflows for FIR-filter equalization of communication signals distorted by intersymbol interference.
- Collected signal measurements and applied FFT-based analysis to compare filter responses, tune coefficients, and improve demodulated-signal recovery.

Sunny Optics

Ningbo, China

Research Intern | Mentor: Chao Xu

May 2018 - Aug 2018

- Designed and optimized automotive LiDAR and mobile-camera lens systems in CodeV/Zemax against optical performance and packaging requirements.
- Worked with opto-mechanical engineers through lab integration, using test feedback to refine alignment and system functionality.

Education

University of Pittsburgh — *Pittsburgh, PA*

Ph.D. in Electrical and Computer Engineering — *Dec. 2026 expected*

Georgia Institute of Technology — *Atlanta, GA*

M.S. in Computer Science, Machine Learning track — *May 2027 expected*

University of Rochester — *Rochester, NY*

M.S. in Optics (thesis track) — *Aug. 2019*

B.S. in Optical Engineering — *May 2017*

B.S. in Applied Mathematics — *May 2017*

- Relevant training: silicon photonics simulation, signal processing, software engineering, deep learning, reinforcement learning, circuit design, lens design, coating design, photonic device design, and computational imaging.

Selected Publications and Talks

Distributed Coherent Optical Computing via Injection-Locked Photonic Networks

Gao, S., K. Lüdge, F. Da Ros, N. Youngblood, submitted to APL, under review, 2026

Distributed Coherent Photonic Computing via Injection Locking

Gao, S., SPIE Optics + Photonics (oral presentation), 2026

Distributed Coherent Optical Computing via Injection-Locked Photonic Networks

Gao, S., Invited PhD Candidate Talk, Pittsburgh Quantum Institute, Pittsburgh, 2025

Strain-free FBG sensors for accurate temperature measurements from cryogenic temperatures to 300 K

Gao, S., P. R. Swinehart, S. Zhong, J. Wuenschell, N. Lalam, M. Buric, K. P. Chen, SPIE Defense + Commercial Sensing (oral presentation), 2023

Preference of subpicosecond laser pulses for terahertz wave generation from liquids

Q. Jin, Y. E., **S. Gao**, X.-C. Zhang, *Advanced Photonics* 2(1), 015001, 2020

Investigation of liquid properties on emitting terahertz wave under ultrashort optical excitation

S. Gao, Q. Jin, Y. E., X.-C. Zhang, SPIE, 2019